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Summary Post

by [Mansour Saeed Mubarak Hamdan Al Hamdani](#) - Friday, 12 September 2025, 11:16 AM

Reflection and Synthesis

In Week 3, I revisited my original post on Agent Communication Languages (ACLs) such as KQML, incorporating peer feedback and insights from Units 1–3. My initial contribution highlighted the standardization and ontology-driven architecture of KQML, which enhances interoperability in multi-agent systems (MAS), while also noting its verbosity, complexity, and security weaknesses compared to method invocation in Python or Java (Labrou et al., 1999). Peer feedback emphasized the semantic clarity of performatives, interoperation issues, and security extensions such as JADE-S, which further illustrated the practical costs (Bellifemine et al., 2001). This aligns with my original concerns regarding implementation overhead.

Peer Agreement and Enhancements

The role of KQML in heterogeneous environments was supported due to its performative richness and scalability (Finin et al., 1992; Zhang et al., 2024). This aligns with Unit 1’s emphasis on coordination. The efficiency concerns I raised were also validated, with references to parsing overhead and the benefits of direct calls in controlled environments (Luo et al., 2012). This reinforced my understanding that choosing between ACLs and method invocation is context-dependent, reflecting one of the central themes of Unit 2. Additionally, proposals to investigate ontologies and large language models (LLMs) (Cheng et al., 2024) added depth to my post by addressing semantics and scalability, key areas of focus in Unit 3.

Personal Growth

Through this process, I developed the ability to balance initial skepticism with a broader perspective, drawing from principles such as protocols of commitment to establish clearer contracts (Singh, 1998). This 300-word summary integrates multiple contributions, improving my original post and connecting classic ACL challenges with modern AI solutions—an essential step for advancing MAS development.

References

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